Vertically Integrated Industries

Investment theses and category views on minerals, metals, manufacturing and chemistry

1. Investment theses: tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and that use their own software and AI to make them faster and with fewer skilled people than older competitors. They are vertically integrated because one company owns the whole job, from order to shipped product, and earns money on what it ships. The five theses below explain why demand is rising, where new companies have an edge, how buyers behave, what the unit economics look like, and where the main risk sits.

1. **The US is moving its supply of minerals, magnets and parts away from China for national security and economic resilience, and defense budgets, government lending and a lasting labor shortage will keep that demand in place for years**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA), and makes about 94% of the world's rare-earth magnets, which go into drones, missiles, motors and robots (Mining.com). The US relies fully on imports for 16 critical minerals (USGS), and China has already used export controls on these materials; its current pause on those controls ends on November 10 and November 27, 2026. Buyers and the government will pay for non-Chinese supply even when it costs more, because supply chain resilience is now an economic and national-security issue.
   - Defense budgets are rising and the Pentagon is paying to rebuild domestic supply of munitions and parts, with the FY27 request asking for $49.5B for 10 key munitions, about 8.5 times the average of the past decade (CNAS), and the Navy putting $900M into submarine-parts plants (Breaking Defense). Because the FY26 defense bill also lets the Pentagon commit to purchases of up to 10 years with advance payments (FY26 NDAA), factories that qualify for defense work can count on years of steady orders.
   - The government now lends to and invests in the companies rebuilding these supply chains, which lowers the cost of building new plants. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to Vulcan Elements, a US magnet maker, and a $500M conditional loan to Phoenix Tailings, a US rare-earth metals maker, and about $20B is requested for it in FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE), and governments worldwide put about $65B into minerals in 2025 (IEA).
   - US manufacturers will stay short of skilled workers for years, since they need to fill about 3.8M jobs by 2033 and about 1.9M of those jobs are projected to go unfilled (Deloitte), which favors plants that can run with fewer people. Of the 17,156 machine shops in the US, 83% have fewer than 20 employees (Census), and many owners are near retirement with no successor.
   - Some of the current tailwinds are temporary, so we should not build the thesis on them. Tariffs are set by proclamation and can change overnight, about 82% of the munitions request depends on a reconciliation bill Congress has not passed (CNAS), and government loans take a long time to close and come with conditions. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR). We should favor companies that still work if tariffs fall and the munitions surge slows.
2. **AI and automation let new companies quote, program and inspect parts faster and get more output from plants that run nonstop, which gives them an edge over older competitors**
   - The work around the machine includes quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. Arda, a factory software startup, says its AI cut the time to write assembly instructions at Anduril from about 10 hours to 15 minutes (claim), and a former Hadrian employee told us in an expert call that Hadrian cut its lead times by about 50% through automation.
   - The machines in most plants sit idle for much of the day, which leaves room to grow output without building anything new. In a survey of more than 100 plants, cutting machines ran about 34% of the time

against a 70% target, and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting), a gap that planning and control software can close without a new plant.

- In a plant that runs nonstop, such as a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case for automation rests on speed and yield more than on headcount.
- Nobody has yet shown that AI makes each unit cheaper as volume grows, so the edge today is speed, yield and the ability to grow with a smaller crew while competitors cannot hire.

3. **Defense and aerospace buyers pay a premium for fast delivery and spare capacity, and a supplier that wins approval keeps the work for decades**
   - Defense buyers care more about delivery dates than price, and a director at a defense prime told us in an expert call that "cost is secondary" because it is "driven usually by schedule", while an engineer there said he would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - A major defense contractor, often called a prime, approves a supplier only after the plant exists and is certified, which takes about nine months to two years (expert calls with two primes). Once approved, a supplier keeps the work for aerospace programs that can run about 50 years.
   - The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that supplier base is small, old and short of people, so primes need new suppliers that can deliver on time.
4. **Companies that own their plants earn the best margins, and the strongest ones get governments, customers or lenders to fund much of the build**
   - Companies that own their plants keep more of each sale than those that route work to partners, and Protolabs, a listed maker of custom parts, earns a 49% gross margin in its own plants and 31% on work it sends to partner shops (company filings). Approved incumbents also earn well, and Howmet Aerospace's engine-parts unit ran a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization) in the second quarter of 2026 (company filings).
   - The central risk is capital spending (capex) on buildings, machines and certifications before contracts are signed, because the money goes out years before revenue comes in. Hadrian, a precision-parts maker, committed about $200M to a single plant in Mesa, Arizona (company release), and its CEO says capex runs about two years ahead of revenue (claim).
   - The companies that succeed shift much of the cost of the build to others, using government loans and grants, purchase agreements signed before the plant is built, factories hosted inside customers' plants, bank debt once revenue exists, leased buildings, franchising, and acquisitions of certified plants.
5. **Established incumbents control supplier approvals and buy promising startups, but an aging and fragmented supplier base still leaves room for new companies**
   - The primes decide which suppliers are approved and how much work each one gets, and they cap each supplier at about a quarter to a third of their work, so even a winning supplier grows one share at a time. Precision Castparts, one of the largest aerospace parts suppliers, saw its share of Boeing's machining work fall from about 50% to 30% over ten years, and it is now automating (expert call).
   - Incumbents often buy new technology instead of building it, which gives startups buyers but also well-funded competitors. BHP's venture arm backs Mariana Minerals, Rio Tinto paid $6.7B for Arcadium Lithium, and RTX Ventures, the venture arm of the defense contractor RTX, led the seed round of Radical AI, an AI materials lab. Drug companies are also building their own AI labs, and Eli Lilly and NVIDIA plan to invest up to $1B over five years in a joint AI lab for drug discovery in the San Francisco Bay Area (Lilly and NVIDIA, January 2026).
   - Hardware companies in this space have failed when their growth ran ahead of their economics, as Markforged, a 3D-printer maker, fell from a $2.1B valuation in 2021 to a $42.5M sale in 2026 (3D Printing Journal), and Daedalus, an AI-run machining startup in Germany, went insolvent in August 2026.

- New companies win where the incumbents are weakest, among small, aging shops with no successor, new defense programs that need second sources, and plants that need software to run with fewer people.

2. Interesting categories

The categories below follow our primer, and for each one we set out the main tailwinds and headwinds and state whether we should focus there, judged on the category as a whole.

A. Raw Materials

Stage A: Locating & finding minerals and mines

- **Tailwinds: Cutting reliance on China starts with finding new deposits of copper, lithium, nickel and rare earths in the US and allied countries, since the US relies fully on imports for 16 critical minerals (USGS).** New mines are needed for batteries, magnets and the grid, and AI combined with cheaper drilling and faster lab testing lowers the cost of checking each target. The major miners are paying for these new methods through their venture arms, which shows that buyers for the work exist.
- **Headwinds: Most exploration projects never become producing mines, and the few that do take 15 to 19 years.** By one estimate only about 0.5% of exploration projects succeed (claim), and the wait is long, since mines average 15.7 years from discovery to production, US mines opened in 2020-24 took 19.1 years, and US permitting alone commonly takes 7-10 years, against about 2 in Australia and Canada.
- **Why we should not focus here: Explorers rarely keep the value of what they find, because the miner that builds the mine takes most of the economics.** Explorers usually bring in a miner through an option or joint venture, in which the miner pays staged cash and shares and funds the drilling to earn 51-70% of the project. The explorer keeps a minority stake or a royalty, usually a net smelter return (NSR) of 1-3% of metal sales after transport, and the miner can often buy part of that royalty back at a fixed price (industry option terms). A royalty pays only once someone else has built the mine and it is producing. An explorer that keeps the mine must raise billions, since a new underground copper mine can cost over $2B (Ecofin), and then wait 15 to 19 years for production. In both cases the return is too small, arrives too late or needs too much capital to justify an early equity investment. Selling exploration software to miners has not solved this, and Earth AI says 80 software pilots it ran with miners did not turn into recurring revenue (claim).

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Stage B: Mining & refining

- **Tailwinds: China controls most of the world's refining and magnet making, so buyers and governments are paying to build that capacity outside China.** China has already used export controls on these materials, which showed buyers the risk of depending on it. Drones, missiles, motors, EVs and data centers all depend on that supply, so defense and auto buyers will pay for a non-Chinese source and sign long purchase contracts to secure it. Pentagon loans and 10-year purchase commitments with price floors add to that demand.
- **The US is rebuilding its own uranium enrichment because a 2024 law bans Russian enriched uranium and new reactors need a fuel that only Russia has sold at commercial scale.** The Prohibiting Russian Uranium Imports Act bans imports of Russian low-enriched uranium (LEU), the fuel for today's reactors, and all waivers end by January 1, 2028 (Public Law 118-62, DOE). Foreign enrichers supplied about 77% of the enrichment bought by US reactor operators in 2025, and Russia alone supplied about 26% (EIA). Most advanced reactor designs need HALEU (high-assay low-enriched uranium, enriched to between 5% and 20%), which no US supplier yet sells and which has been commercially available only from Russia (DOE, Reuters). In January 2026 the Department of Energy awarded $2.7B of contracts to rebuild domestic enrichment, including $900M to General Matter, a startup founded in 2024 that is building a plant on DOE-leased land in Paducah, Kentucky and says it will enrich uranium by the end of the decade (DOE, claim).

- **Headwinds: Western plants take longer and cost more to build than Chinese ones, and they sell against Chinese prices.** Chinese refineries take about 6 months to commission, against 2 to 4 years in the West. A Western plant takes about 5 years to build and another 3 to 5 to ramp after permits (claim), and it costs hundreds of millions of dollars before it earns revenue. Prices for many of these materials are set in China, so a US producer competes against Chinese costs. MP Materials, the largest US rare-earth producer, lost $85.9M on $224.4M of revenue in 2025 (company filings).
- **Why we should focus here: Mining and refining have the most lasting demand in the theme, because the need for non-Chinese metal and magnets comes from national security and supply chain resilience.** Because of that, demand does not depend on a tariff or a single budget cycle. New companies have a real edge here, because restarting old mines that already have permits shortens the timeline, processing scrap and mine waste avoids a new mine altogether, and AI-run process control raises yield in plants that run around the clock. The category is capital-heavy, so the companies that build here pair equity with project debt and long purchase contracts.

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, General Matter, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Metal Supply

- **Tailwinds: Tariffs, buy-American rules and reshoring are raising demand for US metal from machine shops, fabricators and primes.** Job shops want faster quotes and delivery than brokers give them, and online ordering makes that possible.
- **Headwinds: Established distributors already serve this market well and are gaining share without AI, which leaves little room for a new entrant.** The largest is Reliance, Inc., the biggest metals service center company in North America by revenue, with $14.3B of net sales in 2025, about 310 locations and about 17% of all tons sold by US service centers (Reliance 10-K). The second largest is Ryerson, with $4.6B of net sales in 2025, which merged with Olympic Steel in February 2026 to reach about 160 facilities (Ryerson results). Distributors earn a spread on the metal plus cutting fees, and much of their cash sits in inventory, so even at Reliance's scale the pretax margin was 6.8% in 2025, and Ryerson posted a net loss. Online ordering with instant quotes already exists, so technology does little to set a new distributor apart.
- **Why we should not focus here: The margin in metal supply stays with the incumbents, who already have the stock, the buying power and the customers.** A new distributor ties up its capital in inventory and then competes on price and availability, where the incumbents are bigger. The main tailwind is the tariff, which can reverse, and a margin that exists only because of a tariff is not a sound basis for an investment.

Example companies: Nox Metals, Reliance, Ryerson

Fabrication

- **Tailwinds: Reshoring and the growth of US hardware companies are creating demand for metal parts made quickly in the US.** Fabrication means cutting, bending and welding metal into frames and enclosures, and assembling wire harnesses (the bundles of wires that carry power and signals through an aircraft or missile). Demand comes from defense, drones, space and energy hardware companies, and these buyers pay extra for speed. The shortage of welders and machinists favors plants that automate cutting and bending and use software for quoting, nesting (packing parts onto a sheet to cut scrap) and scheduling.
- **Headwinds: The speed premium fades at volume, and US shops beat Chinese prices on only part of their work (claim).** Customers that grow often buy their own machines and take the work in-house.
- **Why we should focus here: Fabrication plants cost far less than precision plants, so companies in this category can grow on their own revenue and reach profit early.** Owned plants keep more of each sale than shops that route work to partners, and software for quoting and scheduling lets a plant handle many

small orders with a small crew. Buyers that pay for speed, low capital needs and early profits make this one of the most attractive categories in the theme.

Example companies: SendCutSend, 1872, Senra Systems, Machina Labs, CircuitHub, Jeevy, Protolabs

Precision & Defense Hardware

- **Tailwinds: The Pentagon needs a larger base of trusted domestic suppliers for precision parts, the metal or composite parts made to very tight limits by machining, 3D printing, casting or molding that go into missiles, jets, satellites and submarines.** Today the Pentagon has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Munitions and submarine budgets are rising, and purchase commitments of up to 10 years are now allowed. Primes outsource much of this work to small shops, and as many owners retire, certified capacity is coming up for sale.
- **Headwinds: A new precision plant has to be built and certified before it wins volume work, so capex runs well ahead of revenue.** AS9100, the aerospace quality standard, typically takes 6-12 months, and approval at each buyer takes about nine months to two years. Much of the munitions money still depends on Congress, and the largest incumbent suppliers are automating.
- **Why we should focus here: An approved supplier keeps the work for programs that can run 50 years, primes pay a premium for fast delivery, and approved incumbents earn strong margins, so the prize at the end of approval is real.** New companies can reach it by building their own plants, which keeps the whole margin but means paying before approval, by buying certified shops from retiring owners at about seven times annual earnings (claim), by running a franchised network in which franchisees buy the machines, or by hosting cells inside a buyer's plant, which shares the cost. Each model trades some margin for less capital, and together they let new companies take share from small, slow shops that cannot keep up with defense demand.

Example companies: Hadrian, Isembard, Divergent, Freeform, Amca, Atomic Industries, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

C. Specialty Chemicals

Producers (hybrid)

- **Tailwinds: Drug companies are moving discovery chemistry out of China because relying on it has become a supply chain, intellectual property and national-security concern.** We limit this category to automated labs that make new molecules for drug programs. BIOSECURE, a law passed in December 2025, limits the use of certain Chinese biotech suppliers, and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO). AI that designs molecules and robots that make them bring down the cost of doing this work in the US and Europe. Chemify, a Glasgow-based lab, says it already works with 8 of the 20 largest drug companies (claim, SCI).
- **Headwinds: US chemists cost several times Chinese rates, so these labs have to automate to reach offshore prices.** The category is young, labs have not disclosed revenue, and drug companies are building their own AI labs. Eli Lilly and NVIDIA plan to invest up to $1B over five years in a joint AI lab for drug discovery (Lilly and NVIDIA, January 2026).
- **Why we should focus here: Drug companies want a domestic or allied source for discovery chemistry, and they now have a legal reason to move work out of China.** These labs make small batches of new molecules for drug teams at milligram-to-gram scale and do not develop their own drugs. They are paid per compound delivered, or through milestones and royalties if a molecule advances. Each lab costs tens of millions to build, but it owns the software and the reaction data, and that data improves its designs with every run. If automation brings unit cost close to offshore prices, the category can take a meaningful share of work that is leaving China.

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Producers

- **Tailwinds: China makes about 85% of battery cathode material and over 98% of the cheaper LFP type, so battery makers for EVs, grid storage and defense need another source for resilience and security.** Data centers are on track to use about 945 TWh of power by 2030, up from 415 TWh in 2024, and cooling takes 7% to over 30% of it, which creates demand for new cooling materials. A new material has typically taken 10 to 20 years and more than $100M to develop (claim), and AI design aims to shorten that.
- **Headwinds: Customers take one to five years to qualify a new material, so revenue arrives slowly even after the material works.** Owned cathode plants cost hundreds of millions of dollars and compete on Chinese prices, and Umicore's battery-materials unit lost money on €436M of revenue (company filings).
- **Why we should focus here: These companies design the material with AI in their own lab, then make it and sell it by the kilogram or ton under supply deals, so the company that designs and makes the material keeps the margin, while contract manufacturing keeps the capital need far below that of a refinery.**

Example companies: Orbital Materials, Mitra Chem, Chemix, Albemarle, Umicore, Corning

Materials Discovery Labs

- **Tailwinds: AI and robotic labs can now make and test new materials far faster than human teams, and the leading labs report making more than a thousand alloys in about six months (claim).** Non-Chinese magnets and battery materials, stronger alloys for defense and better cooling for data centers all require new materials.
- **Headwinds: These labs do not yet sell material at volume, and finding a material faster does not shorten the years customers take to qualify it.** Revenue today comes from paid R&D, licensing, software and government research awards of around $1M, and valuations in the category are high relative to that revenue. Investors note that materials science has not yet had its AlphaFold moment, meaning a single AI breakthrough like the model that predicted protein structures (claim).
- **Why we should focus here, on a longer time horizon: If these labs work, they shorten the slowest part of the materials chain, which is finding a material worth qualifying.** To sell at volume they usually go through a partner that scales the material up and sells it, so a lab earns a fee or royalty unless it builds its own plant.

Example companies: Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Kebotix, Citrine Informatics, Uncountable, Albert Invent, Microsoft Discovery

E. Enabling Software & Robotics

Enablers

- **Tailwinds: Reshoring only works if US plants can run with fewer skilled people, and about 1.9M US manufacturing jobs are projected to go unfilled by 2033 (Deloitte).** The US also lags on automation, with 38,428 industrial robots installed in 2025 against 354,226 in China. Cutting machines in existing plants run only about a third of the time (FTI Consulting).
- **Headwinds: Sales to plants are slow, and incumbent vendors bundle software with hardware they have already installed.** Robots are not yet reliable enough for most factory work, since robot-learning success is 59-66% on short tasks against the 99.9% or more that a factory needs.
- **Why we should focus here: Software and robotics companies do not have to build factories, so their capital goes into product and sales and they can grow faster than plant owners.** Plant-control software can be paid on the extra output it delivers, factory software sells by subscription, and robot makers sell or rent robot cells.

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson